Domesticating the Algorithm: Epistemic Drift and the Erosion of Disciplinary Distinctiveness in Computational Text Analysis

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Abstract

How do the social sciences maintain distinct approaches to knowledge-making when they adopt shared computational methods? This study examines the diffusion of unsupervised topic modeling across five social science disciplines, Sociology, Political Science, Communication, Psychology, and Interdisciplinary Social Sciences, by analyzing the documented implementation decisions in 769 full-text articles published between 2009 and 2024. Using automated large language model extraction validated against human coding, we reconstruct the upstream calibration choices (algorithm selection, K-selection strategy, labeling basis, validation) and downstream analytical deployments (epistemic function, analytical family, link to inferential claims) that constitute the method's documentary practice. Three findings emerge. First, a Multiple Correspondence Analysis of downstream deployment reveals that disciplines occupy systematically distinct positions in the epistemic space: Psychology clusters near a measurement-oriented configuration, Sociology occupies a filtering-oriented quadrant, and Communication anchors a descriptive corner, indicating that the same mathematical architecture is adapted to fit different disciplinary warrant structures. Second, the field is converging on increasingly complex algorithmic architectures (STM, BERTopic), but documented calibration practices are not keeping pace: a significant negative interaction between algorithm complexity and publication year ($b = -0.095$, $p = .009$) indicates that the sophistication advantage of complex models is shrinking over time, a pattern we term epistemic drift. Third, between-discipline distances in the epistemic space are declining (centroid dispersion -64%), though convergence operates asymmetrically: paradigmatically coherent fields consolidate around shared practices, while Sociology fragments internally as some practitioners adopt measurement-oriented approaches while others maintain interpretive positions. Together, these findings demonstrate that computational methods do not standardize scientific practice upon arrival; instead, they produce a dual convergence, technical and epistemic, that may erode the disciplinary diversity that made the method's cross-disciplinary diffusion productive. The study contributes to the sociology of scientific knowledge by providing large-scale quantitative evidence that

the stabilization of underdetermined methods is socially organized through disciplinary evaluative cultures, and that software ecosystems can compress this diversity in ways that are structurally consequential for how the social sciences produce knowledge.

1 Introduction

The rapid expansion of computational text analysis across the social sciences has been widely celebrated as a methodological advance, one that enables the study of cultural, political, and social phenomena at scales previously unattainable (Grimmer & Stewart, 2013; Lazer et al., 2009). Yet the uptake of these tools poses a question that is sociological rather than merely technical: how are methods developed in engineering contexts rendered credible within disciplines organized around different standards of evidence?

Among computational methods, unsupervised topic modeling is a particularly revealing site for this question. Rooted in Latent Dirichlet Allocation (Blei et al., 2003), topic models infer latent structure from word co-occurrence patterns without requiring categories to be defined in advance. A topic model does not produce sociological concepts; it produces probabilistic distributions of words across documents. Meaning emerges only when researchers actively intervene, labeling the topics, justifying the evidentiary basis of those labels, deciding how many topics are sensible, and determining whether the output is sufficiently valid to support substantive claims (Chang et al., 2009; P. DiMaggio et al., 2013). This interpretive negotiation makes the method unusually sensitive to the epistemic culture of the field that adopts it, not because topic modeling is uniquely underdetermined (many methods require consequential judgment), but because the discretionary decisions are especially visible, recoverable from published texts, and variable across disciplines.

This visibility is analytically important. Many methods involve consequential judgment, but the documentary record does not always preserve those judgments in a comparable form. Topic modeling papers often document, or fail to document, precisely the decisions through which latent statistical structure is translated into a substantive claim. The method therefore allows us to study not only adoption, but the visible stabilization of methodological credibility: what authors make explicit, what they leave tacit, and what reviewers are consequently able to evaluate.

When topic modeling diffused across the social sciences, it did not impose a single analytic logic. Rather, it functioned as a boundary object (Star & Griesemer, 1989): robust enough to maintain a shared mathematical identity, yet plastic enough to be adapted to distinct local needs. Existing scholarship has established that disciplines use computational text methods for different substantive purposes. But we still lack systematic evidence on how disciplines differ in the concrete implementation decisions, the selection of K, the basis for labeling, and the choice of validation procedure, through which topic models are stabilized as credible findings over time.

Two trajectories are possible. Disciplines may continue to stabilize the method according to their distinct local norms, achieving what Timmermans and Berg (1997) call "local universality." Alternatively, the default configurations of software packages may be driving practice toward uniformity across disciplinary boundaries (Akrich, 1992). These trajectories are not mutually exclusive; our data cannot isolate the causal mechanism behind either. What we can test is the structural pattern: whether architectural convergence is accompanied by a corresponding deepening of calibration, or whether the two decouple, producing what we term epistemic drift.

To investigate these questions, this study systematically analyzes the documentary record of published topic modeling research. Drawing on an automated, large language model-driven extraction of 769 full-text articles published between 2009 and 2024, we reconstruct the methodological decisions that researchers make when fitting, calibrating, and deploying topic models across five social science disciplines: Sociology, Political Science, Communication, Psychology, and Interdisciplinary Social Sciences. We address three interconnected research questions, each building on the last:

RQ1: How do distinct social science disciplines differ in the documented implementation decisions through which topic models are stabilized? We test whether disciplinary differences extend from downstream deployment to upstream calibration choices.

RQ2: Is the field converging on complex algorithmic architectures over time, and is this convergence decoupled from documented methodological calibration?

RQ3: Is the between-discipline distinctiveness in downstream deployment eroding over time, and is this contraction consistent with architectural standardization or with the healthy maturation of a shared research frontier?

Together, these questions address a broader sociological problem: how evaluative standards travel across fields and how disciplinary autonomy is maintained or eroded under conditions of methodological uncertainty. This problem extends beyond topic modeling to the ongoing adoption of computational and generative AI methods across the social sciences.

2 Literature Review

2.1 Topic Modeling as an Interpretively Underdetermined Method

The credibility of a topic model depends on decisions for which the field has not converged on a single accepted rule. Chang et al. (2009) demonstrated that

models judged mathematically superior by standard metrics frequently yield topics that human readers find less coherent, decoupling internal model fit from substantive interpretability. Proposals for selecting the number of topics range from density-based measures to heuristic procedures (Arun et al., 2010; Cao et al., 2009). Automated coherence metrics yield different evaluative outputs depending on construction choices and fail to reproduce domain-expert judgment reliably (Mimno et al., 2011; Röder et al., 2015). A recent review of 789 applied studies confirms this pluralism, documenting heterogeneous validation practices and the absence of any dominant validation protocol across computational social science (Bernhard-Harrer et al., 2025).

A comparison with regression clarifies the stakes. Regression rests on an institutionalized apparatus of identification assumptions and diagnostic tests that structure execution and critique across disciplines (Angrist & Pischke, 2010; Freedman, 1991). Topic modeling lacks any comparably institutionalized validity apparatus. Drawing on Espeland and Stevens (1998), we can understand this absence as a structural condition: commensuration, the transformation of different qualities into a shared evaluative metric, has not been achieved for topic modeling. Attempts to impose it through coherence scores or mechanical reproduction protocols carry evaluative assumptions that are not epistemologically neutral (Bowker & Star, 1999). In the absence of successful commensuration, each disciplinary community must develop its own standards for what constitutes a credible topic model. The stabilization of topic modeling is, in this sense, an empirical question about how discretionary judgment is rendered defensible, and by whom.

2.2 Disciplinary Domestication of a Shared Method

If credibility depends on locally organized stabilization, then different academic communities should resolve the underdetermination differently. Disciplines operate with distinct warrant structures, configurations of evidence, proof, and evaluative criteria that determine what counts as a defensible claim (Becher & Trowler, 2001; Knorr Cetina, 1999). Star and Griesemer (1989)'s concept of boundary objects explains how a single artifact can serve these different communities simultaneously: the mathematical core is invariant, but the epistemic object that emerges, what the model is taken to show, varies with the evaluative norms of the adopting field (Callon, 1986; Timmermans & Berg, 1997).

Discipline-specific literatures suggest what form this variation should take. Sociology has emphasized inductive cultural mapping, preserving the analyst's authority to interpret and reinterpret algorithmic output (Bail, 2014; P. DiMaggio et al., 2013). Political science integrated topic modeling into inferential frameworks, with the Structural Topic Model providing a pipeline for translating topic distributions into controlled variables (Grimmer & Stewart, 2013; Roberts et al., 2014). Communication has emphasized descriptive content mapping at scale (Maier et al., 2018). Many applications in psychology treat topic distributions as measurement proxies for latent constructs (Eichstaedt et al., 2015; Kjell et al., 2019). These contrasts are recognized, but they have not

been tested comparatively against the full range of upstream and downstream implementation decisions that constitute the method's documentary record.

If topic modeling functions as a boundary object, disciplinary differences should appear not only in downstream deployment but also in upstream calibration: how K is selected, what evidentiary basis is used for labeling, and whether formal validation is conducted. Discipline is unlikely to be the only source of variation. Corpus type, journal conventions, co-authorship with computational specialists, and graduate training may also shape implementation choices. Our claim is narrower: if topic modeling is domesticated through local warrant structures, discipline should remain a patterned axis of variation even amid these other influences. That is the gap our first analysis addresses.

2.3 Software Standardization and the Calibration Gap

While disciplinary warrant structures drive diversification, software ecosystems introduce countervailing pressures. Toolchains do not merely implement methods; they configure what competent practice looks like by embedding assumptions about default parameters, diagnostic outputs, and the appropriate sequence of operations (Akrich, 1992). The `stm` R package, for instance, scripts a repeatable workflow, estimation, metadata integration, and visualization, that researchers can follow without calibrating the mathematical decisions embedded in the pipeline to the characteristics of their corpus (Roberts et al., 2019).

The consequences of routine defaults are not trivial. Preprocessing choices, stopword removal, and vectorization methods can substantially alter topic structures and interpretability (Denny & Spirling, 2018; Schofield et al., 2017). If researchers adopt defaults to achieve reviewer legibility without tuning parameters to their analytical problem, the resulting convergence may reflect the architecture of the package rather than the structure of the data. Under conditions of methodological uncertainty, where no settled evaluative standard exists, the adoption of the most technically current architecture may be read by reviewers as a signal of competence, consistent with mimetic isomorphism under uncertainty (P. J. DiMaggio & Powell, 1983). Alternative explanations for convergence, genuine technical improvement, expansion of the user base, and growing availability of tutorials, also deserve acknowledgment; our data cannot fully adjudicate among them.

This framework generates two empirical predictions. If the field is experiencing epistemic drift, we should observe architectural convergence decoupled from documented calibration: complexity increasing while calibration remains flat. And if standardization pressures are strong enough to override disciplinary domestication, between-discipline distances in the space of downstream deployment should be shrinking over time. Convergence is not inherently pathological. If disciplines were converging on shared practices of validation, calibration, and evidentiary justification, this would indicate the healthy maturation of a methodological frontier. The concern arises when convergence occurs primarily at the level of software architecture and reviewer-legible workflow, while the discretionary practices that make those tools substantively credible remain thin

or unevenly documented. Convergence that reflects genuine learning would constitute healthy maturation; convergence associated with architectural standardization, without corresponding deepening of calibration, would suggest the erosion of what we term productive plasticity: the capacity of the boundary object to serve diverse epistemic communities. Whether the data support this interpretation is what our second and third analyses test.

3 Data and Methods

3.0.1 Corpus Construction and Bibliometric Bounding

To empirically audit the domestication of computational text analysis in the social sciences, we constructed a bibliometric corpus of research articles published between 2009 and 2024. This interval captures the field's diffusion phase, from early programmatic statements that articulated computational social science as a distinct methodological agenda (Lazer et al., 2009), through the stabilization of accessible software infrastructures (Roberts et al., 2014), to the recent proliferation of neural and embedding-based architectures that further lowered the barrier to entry for large-scale text analysis (Kozlowski et al., 2019).

We retrieved records from the Web of Science (WoS) Core Collection, a curated index of peer-reviewed scholarly outputs that provides standardized bibliographic metadata (e.g., journal, year, authorship, citations) and consistent disciplinary categorization. WoS was selected because it supports reproducible corpus construction at scale: the combination of stable indexing practices, metadata completeness, and disciplinary labels enables systematic comparisons across fields and over time. We implemented a lexical search strategy targeting titles, abstracts, and author keywords. To capture the probabilistic mixed-membership lineage of topic modeling while minimizing false positives, we used the following query string:

("topic model" OR "latent dirichlet allocation" OR "structural topic model*" OR "dynamic topic model*" OR "correlated topic model*" OR "BERTopic" OR "contextualized topic model*" OR "probabilistic latent semantic analysis")*

The included terms serve two functions. First, they cover canonical model families commonly recognized as "topic modeling" in the computational text analysis tradition (e.g., LDA and its extensions; structural, dynamic, and correlated variants). Second, they capture more recent developments that are frequently framed as topic modeling in contemporary applied work (e.g., BERTopic; contextualized topic models). At the same time, the string is intentionally conservative. Notably, we excluded the standalone acronym "LDA" because, in several areas of the behavioral science literature, it often refers to Linear Discriminant Analysis, a supervised classification technique unrelated to topic modeling, which would inflate false positives.

We restricted results to English-language articles (excluding book reviews, editorials, and proceedings) indexed in five WoS categories central to our comparative design: Sociology, Political Science, Communication, Psychology

(including Applied, Experimental, Social, and Interdisciplinary), and Social Sciences Interdisciplinary. The initial retrieval yielded a raw corpus of $N = 1047$ records. The disciplinary distribution already indicates substantial heterogeneity in adoption: Communication ($n = 330$), Psychology ($n = 212$), and Political Science ($n = 187$) constitute core adoption clusters, followed by Social Sciences Interdisciplinary ($n = 180$) and Sociology ($n = 140$). To protect the integrity of the unit of analysis, we applied a multi-stage deduplication protocol designed to identify cross-indexed and repeated entries. We matched records using WoS unique identifiers (UT), digital object identifiers (DI), and normalized titles (TI). This process removed 77 duplicates, producing a final bibliometric corpus of $N = 970$ unique research articles.

3.0.2 Full-text Retrieval

Traditional bibliometrics, which rely solely on metadata, can measure scientific visibility, but they cannot measure the situated methodological work through which models are rendered credible. Abstract summaries rarely disclose the mundane technical choices necessary to answer our questions regarding algorithmic monoculture (RQ2) and methodological opacity (RQ3), such as how topic number was selected or whether topics were validated by human coders. Therefore, we constructed a full-text corpus directly linked to our bibliometric baseline. Unique identifiers (UT, DOI) served as primary keys connecting WoS metadata to PDF files. Full texts were retrieved primarily via the Zotero API. However, automated retrieval pipelines systematically overrepresent open-access materials. Because open-access funding is highly correlated with institutional prestige and specific STEM-adjacent grants, relying solely on API retrieval introduces an "accessibility bias" that would artificially skew our test of scientific capital and journal quartile (RQ3). To mitigate this, we supplemented the automated pipeline with a rigorous manual acquisition protocol for missing items. From the target population of $N = 970$ articles, we successfully acquired and verified a final analytical corpus of $N = 769$ full-text PDFs.

3.1 Full-Text Parsing and Structural Classification

Traditional bibliometrics measure scientific visibility, but they cannot measure the situated methodological work through which computational models are rendered credible. To extract the epistemic justifications and technical minutiae embedded within the text, we developed an automated pipeline utilizing `gemini-2.5-pro`, a large language model (LLM) capable of high-resolution semantic extraction. However, because the structural formatting of scholarly articles varies widely, passing raw PDFs to an LLM risks severe context dilution and hallucination.

To resolve this, the pipeline executes a three-stage parsing and extraction sequence. First, we converted the unstructured PDF articles into structured TEI XML using GROBID (GeneRation Of BIbliographic Data). Unlike simple optical character recognition (OCR), GROBID is a machine-learning library

relying on conditional random fields and deep learning sequence models that are intrinsically layout- and position-aware. It achieves state-of-the-art benchmark performance (F1 scores consistently larger than 0.90) for extracting document hierarchy, reliably differentiating body paragraphs from headers, footnotes, and bibliographic references. This structural precision is crucial to prevent our downstream LLM extraction from ingesting irrelevant noise.

Second, because section headers reflect highly localized, discipline-specific writing conventions rather than a universal standard, we implemented a hybrid classification procedure to map authors' idiosyncratic headers into six canonical categories: Introduction, Literature Review, Method, Results, Discussion, and Conclusion. We first applied a rule-based heuristic using keyword matching. For headers resisting heuristic classification, we deployed an LLM classifier. To maximize accuracy without overloading the prompt, we fed the LLM precise contextual features: the raw header string, its absolute location (section index), its relative location (preceding and succeeding headers), and representative boundary sentences from the section's head and tail. The model returned a classification alongside a confidence score; any assignment falling below a 0.3 confidence threshold was flagged and subjected to manual human validation.

3.2 Targeted Text Bundling and Automated Extraction

With the corpus structurally standardized, we executed the semantic extraction. Rather than submitting entire articles to a single prompt, we dynamically routed specific extraction schemas to targeted "text bundles." This design choice reflects the empirical reality of computational research: methodological decisions are rarely neatly confined to a "Methods" section. We constructed two primary analytical bundles. First, the Technique Bundle concatenated the Method and Results sections. This bundle was queried for the operational details of the models (e.g., algorithm choice, hyperparameter tuning). Including the Results section here was theoretically paramount; critical decisions, such as K-selection heuristics or the human-in-the-loop topic-labeling process, are frequently justified post hoc as analysts evaluate their initial outputs. Second, the Downstream Task Bundle concatenated the Results, Discussion, and Conclusion sections. This bundle was queried to extract the overarching epistemic functions of the topic-modeling output, allowing us to accurately measure disciplinary domestication (RQ1). The complete extraction schemas and exact prompt phrasing are provided in the methodological appendix.

3.2.1 Extraction Schema and Variable Definitions

The automated pipeline extracted two families of variables from each article, corresponding to the two text bundles described above. Because the operationalization of these variables is central to the analyses that follow, we describe each family in turn. Complete extraction schemas and exact prompt specifications are provided in the methodological appendix.

Upstream technical variables capture the operational decisions researchers make when fitting a topic model. Algorithm codes the primary model used (LDA, STM, BERTopic, NMF, Other, or NOT_REPORTED). K-selection strategy codes how the final number of topics was determined, from theory-driven priors through named quantitative metrics to human judgment. Labeling basis codes the evidentiary foundation for topic labels (top words only, top words plus documents, FREX-based, or theory-driven). Additional fields capture hyperparameter reporting, covariate use, preprocessing documentation, label validation, and replication material sharing. Each field was coded using a precision-first protocol: a variable was marked present only if explicit textual evidence triggered the schema; otherwise it defaulted to NOT_REPORTED.

Downstream analytical variables capture how researchers deploy the topic-model output. Because a single article may use output for multiple purposes, the schema extracted up to five discrete components per article. For each component, we coded Epistemic Function (Descriptive Mapping, Data Reduction, Measurement Tool, Classification Proxy, or Exploratory for Theory), TM Link Type (Structure, Variables, or Filter), and Analytical Family (Descriptive Presentation, Statistical Modeling, Computational Analysis, or Qualitative Engagement). These variables operationalize the research question by capturing not only what researchers do with the output but also how they connect it to their substantive claims, the site where disciplinary warrant structures become empirically visible. Component-level variables were aggregated to the paper level using dominant-value assignment for cross-tabulations and binary indicators for geometric methods.

3.2.2 Variable Recoding

Sparse categories ($n < 30$) were merged with theoretically proximate modalities: Classification Proxy with Measurement Tool, Exploratory for Theory with Descriptive Mapping, and Network Analysis with Computational Analysis. For upstream variables, NOT_REPORTED represents a genuine reporting choice. For downstream variables, missingness arises structurally: 158 papers lacked codeable downstream analysis, and 13 additional papers had ambiguous extraction. The MCA and convergence analyses restrict the sample to $N = 598$ papers with complete downstream profiles, treating excluded papers as a structurally distinct population rather than as randomly missing data.

3.3 Analytic Strategy

RQ1: We assess disciplinary variation using bootstrapped, cluster-robust chi-square tests (1,000 iterations, clustered by DOI to account for 85 cross-domain duplicates) and a Multiple Correspondence Analysis on $N = 598$ papers. The MCA uses three active variables (Epistemic Function, TM Link Type, and Analytical Family) with discipline as a supplementary variable, ensuring that the geometric space is structured by documented practice rather than disciplinary labels. Upstream variables are excluded from the MCA because

preliminary analyses showed that they clustered near the origin and diluted explained variance without altering the dimensional interpretation.

RQ2: We construct two paper-level measures: an ordinal Algorithm Complexity score (0 = LDA/NMF, 1 = STM/specialized, 2 = BERTopic) and a Methodological Sophistication Index (0–5) summing five binary calibration indicators (hyperparameter reporting, structural covariates, named-metric K-selection, preprocessing documentation, and formal label validation). This index measures documented calibration depth, not calibration that may have been conducted but left unreported. The decoupling hypothesis is tested via an interaction model: Sophistication $\sim$ Complexity $\times$ Centered Year + Discipline, with cluster-robust standard errors. A negative interaction indicates that the sophistication advantage of complex models is shrinking over time.

RQ3: We compute discipline-level centroids in the MCA space for two periods (2009–2019 and 2020–2024), measure pairwise Euclidean distances, and test the year-level correlation between time and between-discipline centroid variance. A paper-level regression of distance from the field centroid on centered year, discipline, and their interaction provides complementary evidence. Within-discipline spread (trace of the covariance matrix) is compared between periods to distinguish uniform convergence from internal fragmentation.

3.4 Validation

Because of the inherent limitations of LLM-based extraction, including the risk of hallucination and context dilution, we adopted a "precision-first" extraction philosophy throughout the pipeline. The LLM was instructed to code a methodological step as present only if explicit textual evidence, supported by verbatim quotes, triggered the schema. If the required phrasing was absent, the variable defaulted to NOT_REPORTED. This conservative design prioritizes measurement validity over coverage. By strictly measuring what is explicitly justified in the text versus what is left tacit, we operationalized methodological opacity as an observable outcome of reporting practice rather than as an artifact of extraction failure. To formally audit the pipeline, a random sample of 20 articles per domain was drawn from the corpus and manually coded by human raters across the core methodological variables. We calculated Cohen's kappa to evaluate inter-rater reliability between the human coders and the LLM pipeline. The pipeline achieved a Cohen's kappa exceeding 0.85 across all analytical fields, peaking at near-perfect agreement ($\kappa = 0.99$) for algorithm choice.

4 Results

4.1 Descriptive Overview of Computational Practice

The analytic sample consists of $N = 769$ articles published between 2009 and 2024, distributed across Communication (25.4%), Psychology (22.1%), Interdisciplinary Social Sciences (19.5%), Political Science (18.6%), and Sociology (14.4%).

The distribution confirms that topic modeling has diffused broadly across the targeted domains, but not evenly: Communication and Psychology account for nearly half of the corpus, while Sociology remains the smallest disciplinary cluster.

Algorithmic choices also vary substantially. LDA remains the most frequently reported algorithm (50.2%), followed by STM (27.4%) and a residual category including BERTopic, NMF, and other specialized architectures (22.4%). The field, in other words, is no longer organized around a single dominant architecture, even if LDA continues to occupy the largest share of applications. Documented transparency is limited. Across the corpus, 42.4% of articles fail to disclose at least two of three core procedural parameters: the software package used, the strategy for selecting K, or hyperparameter settings. In addition, 86.5% report no formal label validation, and 88.8% provide no replication code or data. These frequencies matter because the analyses that follow concern the documentary stabilization of computational practice rather than practice in the abstract. At baseline, the published record suggests a field in which core modeling decisions are often left only partially visible. (See Table 1 for full descriptive characteristics.)

4.2 Disciplinary Variation in Implementation Practice

Our first question is whether disciplines differ in how they implement a shared computational architecture. If topic modeling had diffused as a standardized technical upgrade, implementation choices should be largely uniform across fields. Instead, the evidence points to patterned disciplinary variation.

Bootstrapped, cluster-robust chi-square tests (1,000 iterations, clustered by DOI) show statistically significant disciplinary variation across all implementation dimensions tested: algorithm choice, K-selection strategy, labeling basis, epistemic function, and analytical family (all $p < .05$; full results, including Cramér's V, are reported in Appendix Table A1). The importance of this result is not simply that disciplines use topic models differently downstream. It is that disciplinary differentiation extends upstream, to the discretionary choices through which the model is rendered stable and defensible in the first place. K-selection and labeling are not merely technical preliminaries; they are among the principal sites where disciplinary standards become visible in the published record.

To assess whether these differences form coherent configurations rather than isolated associations, we estimated a Multiple Correspondence Analysis (MCA) on $N = 598$ papers with codeable downstream analysis. The analysis uses three active variables, Epistemic Function, TM Link Type, and Analytical Family, with discipline treated as a supplementary variable. This design matters substantively: the geometry is structured by documented methodological practice, while disciplines are projected into that space as distributions of practice rather than as defining inputs. The MCA therefore complements the chi-square tests by showing not only that disciplines differ, but also how those differences cluster into distinct multivariate configurations.

Table 1. Descriptive Characteristics of the Corpus (N = 769)

Variable	N	%
Panel A: Discipline		
Sociology	111	14.4
Political Science	143	18.6
Communication	195	25.4
Psychology	170	22.1
Interdisciplinary SS	150	19.5
Panel B: Algorithm		
LDA	386	50.2
STM	211	27.4
BERTopic	26	3.4
NMF	17	2.2
Other/Specialized	87	11.3
NOT_REPORTED	42	5.5
Panel C: Software (top 5 + other)		
stm (R)	137	17.8
topicmodels (R)	52	6.8
MALLET	50	6.5
Gensim (Python)	57	7.4
BERTopic (Python)	20	2.6
Other	79	10.3
NOT_REPORTED	350	45.5
Panel D: K-Selection Strategy		
Named Metric Plus Human	225	29.3
Named Metric Only	160	20.8
Human Judgment Only	140	18.2
Heuristic Or Rule Of Thumb	50	6.5
Theory Or Design Prior	25	3.3
NOT REPORTED	169	22.0
Panel E: Transparency Dimensions		
Reporting opacity (≥2 undisclosed)	326	42.4
K-selection NOT REPORTED	169	22.0
Software NOT REPORTED	350	45.5
Hyperparameters NOT REPORTED	556	72.3
No label validation reported	665	86.5
No replication materials shared	683	88.8

Note. N = 769 papers across five social science disciplines, 2009–2024.

Table 1: Descriptive characteristics of the analytic corpus and the visibility of key reporting practices.

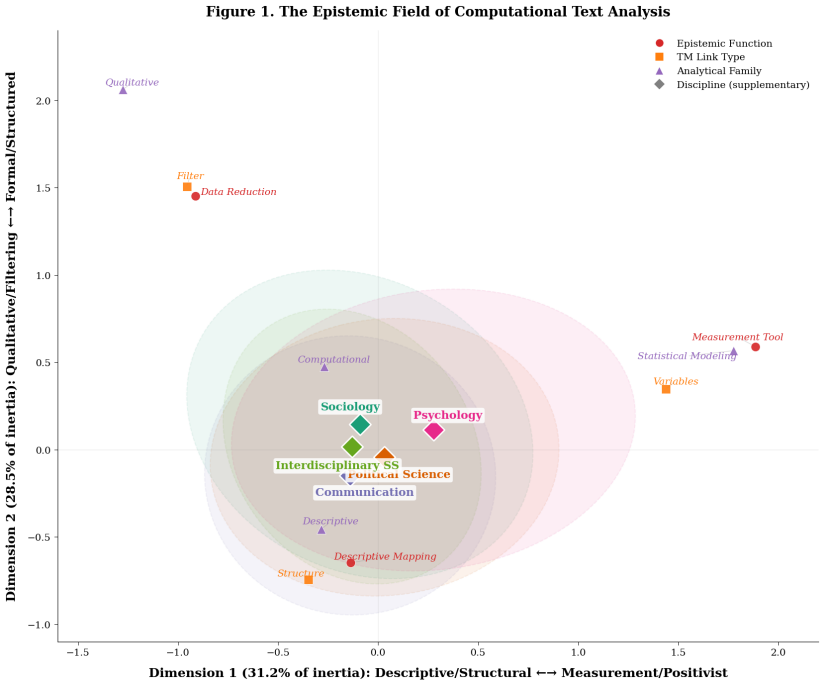

Figure 1: Multiple Correspondence Analysis of downstream deployment

The first two dimensions account for 59.7% of total inertia. Because raw MCA inertia is not directly equivalent to explained variance in linear models, this figure should be interpreted cautiously (**GreenacreBlasius2006**). Here it reflects the relatively tight structural coherence of three closely related categorical variables rather than an unusually strong model in a linear sense.

Dimension 1 (31.2%) differentiates a measurement-oriented configuration from a filtering-oriented one. On the positive end, Measurement Tool, Variables, and Statistical Modeling cluster together, marking papers that treat topic-model output as a quantitative device within inferential workflows. On the negative end, Qualitative Engagement, Filter, and Data Reduction cluster together, marking papers that use the model to reduce textual complexity or retrieve documents for human interpretation. Dimension 2 (28.5%) differentiates active filtering from descriptive mapping. On the positive end, Qualitative Engagement and Filter lie apart from both the measurement cluster and the descriptive center, capturing a mode of use in which researchers actively intervene in algorithmic output to select, reduce, or interrogate the corpus. On the negative end, Structure, Descriptive Presentation, and Descriptive Mapping converge, characterizing papers that present the topical organization of a corpus as a finding in itself.

Disciplines occupy distinct positions in this space. Psychology (+0.276 on Dimension 1) lies closest to the measurement-oriented region. Political Science (+0.033) is positioned near the centroid, spanning measurement and descriptive applications rather than occupying a single pole. Sociology (−0.091, +0.145) occupies the filtering quadrant, while Communication (−0.140, −0.148) anchors the descriptive corner. The same mathematical architecture is therefore being inserted into different epistemic workflows depending on the discipline in which it is deployed.

Within-discipline spreads also vary. Communication exhibits the tightest spread (trace = 1.168), indicating a relatively consolidated range of downstream practice. Psychology shows the widest (trace = 1.670), consistent with heterogeneity across social, clinical, and experimental subfields. Sociology occupies an intermediate position (trace = 1.529). These patterns suggest that domestication is not merely a between-discipline process; it also varies with the degree of internal paradigmatic coherence within a field.

4.3 Methodological Convergence Over Time

Our second question concerns the field's longitudinal trajectory. We ask whether documented implementation practices are converging over time, and whether any architectural convergence is accompanied by a corresponding increase in documented calibration depth.

We operationalized this in two ways. First, we constructed an ordinal Algorithm Complexity score (0 = LDA/NMF; 1 = STM/specialized architectures; 2 = BERTopic). Second, we constructed a Methodological Sophistication Index (0–5) capturing documented calibration depth across five binary dimensions. This index measures what is made visible in the published article, not the full calibration work that may have occurred offstage.

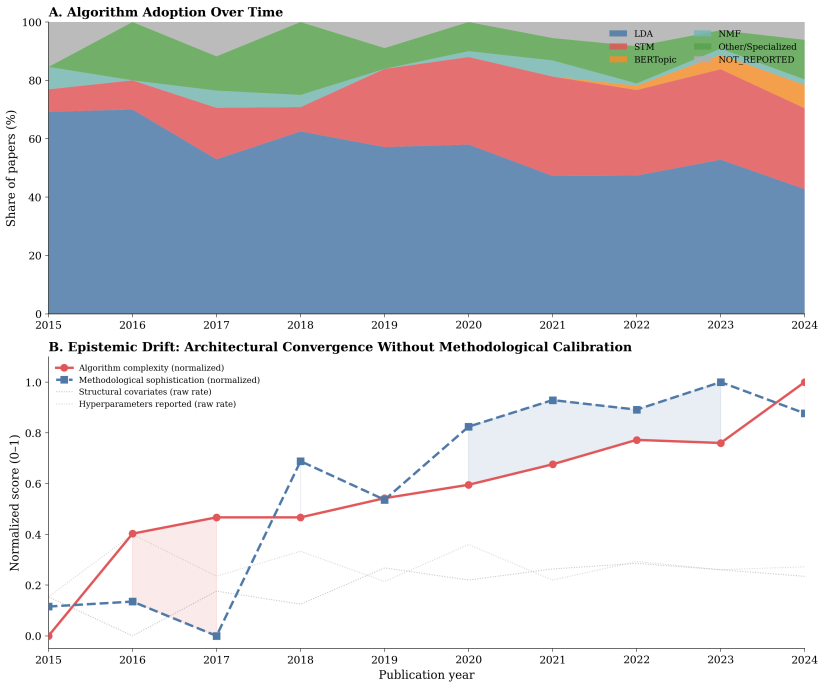

Figure 2: Methodological convergence over time. Panel A: changing algorithm adoption; Panel B: the divergence between algorithm complexity and documented methodological sophistication.

Logistic regressions using centered year (centered at 2019) confirm that LDA use is declining (OR = 0.904, $p = .003$), while STM is rising (OR = 1.083, $p = .043$) and BERTopic is increasing sharply (OR = 2.578, $p < .001$). Mean algorithm complexity also rises significantly over time ($b = 0.044$/year, $p < .001$). At the architectural level, the field is clearly moving away from foundational models toward more integrated and computationally elaborate ones.

That shift, however, is not matched by a broad increase in documented calibration. Of the five sophistication components, only named-metric K-selection increases significantly (OR = 1.200, $p < .001$). Hyperparameter reporting, covariate use, and label validation all exhibit flat trajectories. The overall sophistication index rises modestly, but that increase is carried largely by one component rather than by a general deepening of documented methodological engagement.

A regression of sophistication on complexity, centered year, and their interaction yields the central result. Complexity is positively associated with sophistication at baseline ($b = +0.618$, $p < .001$), but the interaction is negative and significant ($b = -0.095$, $p = .009$ in the full model with discipline controls). The sophistication advantage associated with complex architectures is therefore shrinking over time. Figure 2 visualizes this decoupling: complexity and sophistication rise together through roughly 2019, but diverge after 2020, with architectural complexity continuing upward while documented sophistication plateaus. These trends concern the documentary record rather than all calibration work undertaken by researchers, but they nonetheless indicate that the visible justificatory work accompanying complex models is not increasing in parallel with adoption.

4.4 Epistemic Convergence and Disciplinary Distinctiveness

Our third question asks whether the disciplinary distinctiveness documented above is itself changing over time. To evaluate this, we projected all 598 papers into the MCA space and computed discipline-level centroids for two periods: 2009–2019 ($n = 111$) and 2020–2024 ($n = 487$).

The centroid analysis reveals a pattern consistent with convergence. Seven of ten disciplinary pairs moved closer together. Mean pairwise centroid distance declined by 32.6% ($0.426 \rightarrow 0.287$), and total centroid dispersion fell by 64% ($0.122 \rightarrow 0.044$). Psychology showed the largest movement, shifting from the most measurement-oriented position in the early period (Dim1 = +0.593) toward the center (+0.212). Sociology also shifted, moving from the descriptive region (−0.210, +0.009) toward the filtering quadrant (−0.057, +0.184).

Year-level evidence points in the same direction. The correlation between time and between-discipline variance is $r = -0.76$, $p = .049$. Because this estimate is based on only seven year-points, it should be interpreted as supporting rather than definitive evidence. A paper-level regression provides complementary support: the baseline year effect is negative ($b = -0.036$, $p = .037$), while the

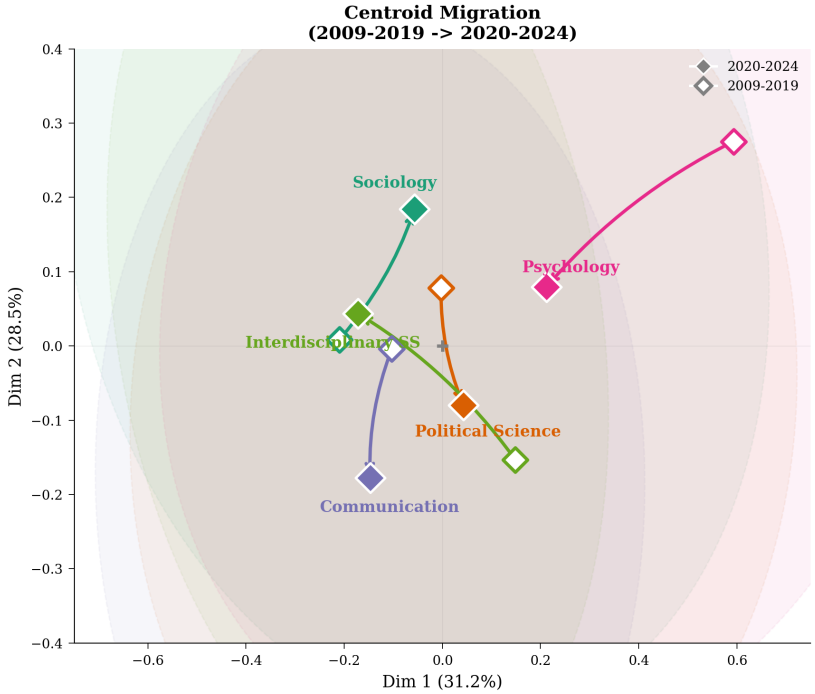

Figure 3: Centroid migration in the MCA space by discipline, 2009–2019 versus 2020–2024.

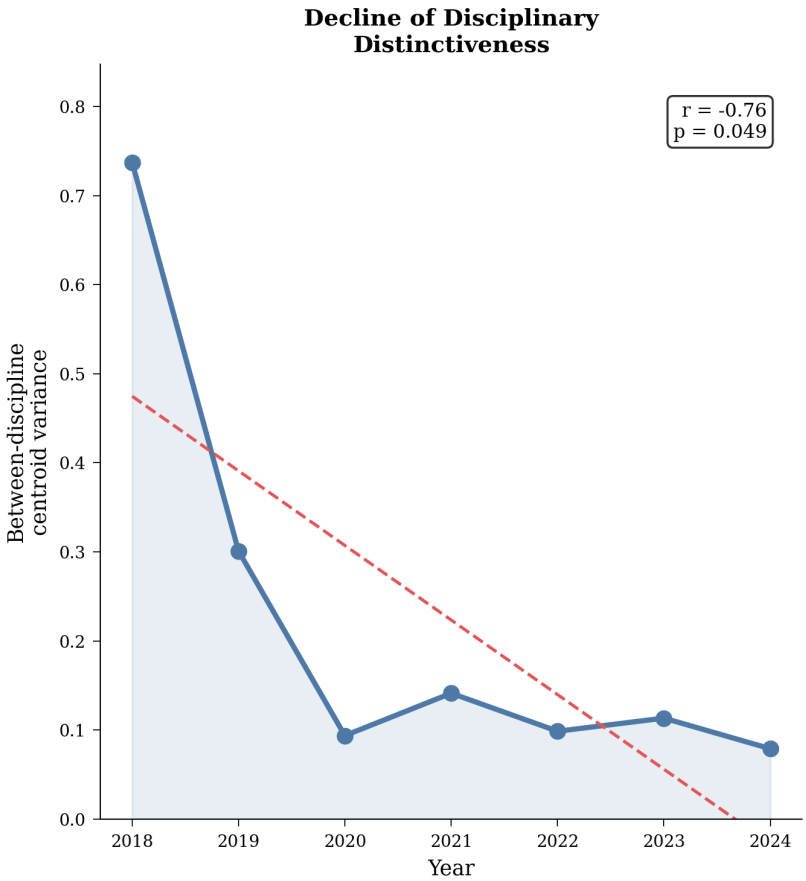

Figure 4: Year-level between-discipline variance in MCA space

Sociology × Year interaction is positive and significant ($b = +0.063$, $p = .043$), indicating that Sociology is not converging uniformly toward the field center.

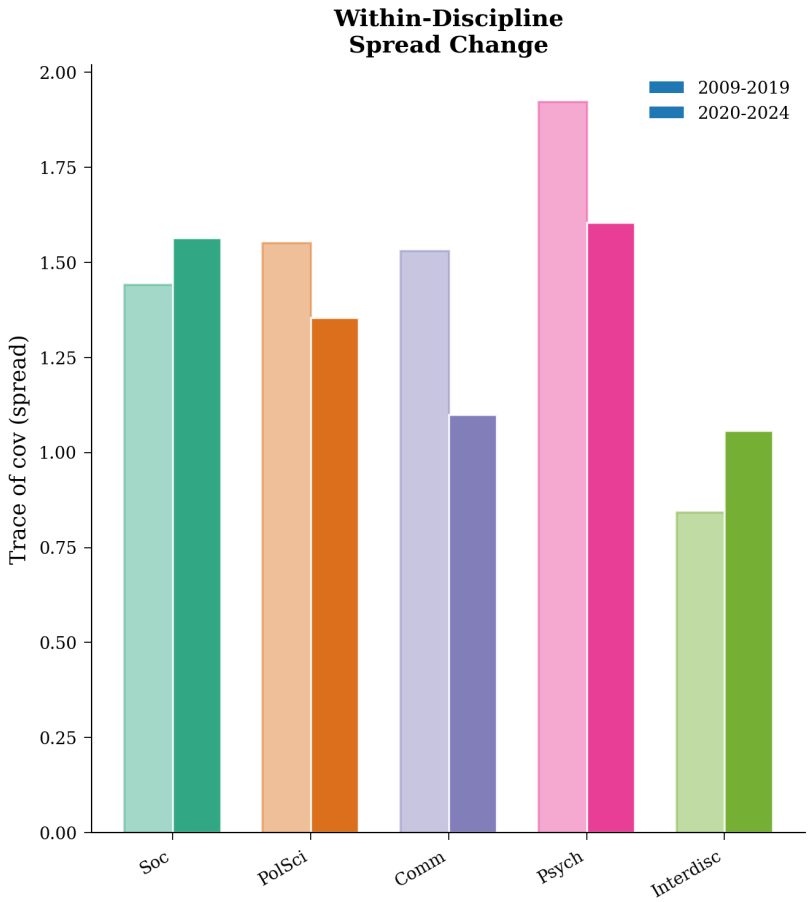

Figure 5: Within-discipline spread in early and late periods

The within-discipline spread analysis clarifies the asymmetry. Psychology, Communication, and Political Science exhibit narrowing spreads over time, consistent with consolidation around shared practices. Sociology and Interdisciplinary Social Sciences, by contrast, show widening spreads, indicating fragmentation rather than uniform consolidation. The underlying indicators are consistent with this reading: between periods, Sociology's Measurement Tool rate rose from 10.0% to 14.5%, and its Variables rate nearly doubled (15.0% → 27.5%), while Psychology's corresponding rates declined sharply. The result is not a simple replacement of one disciplinary logic by another, but a contraction in between-discipline distance combined with uneven within-discipline reorganization.

5 Discussion

These findings compose a cumulative empirical account whose implications extend beyond the specific case of topic modeling. Disciplines domesticate the same algorithm into distinct epistemic configurations (RQ1). Over time, the field converges on increasingly complex architectures without a corresponding deepening of documented calibration (RQ2). And the disciplinary distinctiveness documented in RQ1 is itself eroding, though not uniformly, as formerly extreme positions retreat toward a shared center (RQ3). The significance of the paper therefore lies not only in documenting variation in method use, but also in identifying a historically specific pattern in how computational methods are stabilized, justified, and standardized across the social sciences.

The MCA geometry provides unusually precise evidence that topic modeling operates as a boundary object in the sense described by Star and Griesemer (1989). The mathematical architecture is shared, but the epistemic object that results, what the model is taken to demonstrate, varies systematically with the warrant structures of the adopting discipline. Psychology's proximity to the measurement pole, Sociology's location in the filtering quadrant, and Communication's anchoring of the descriptive corner are not random; they map onto the evaluative traditions that Knorr Cetina (1999) would recognize as distinct machineries of knowing. The finding that upstream decisions, K-selection strategy and labeling basis, also vary by discipline is particularly significant, because it indicates that domestication extends beyond what researchers do with the output to how they stabilize the model itself. The discretionary decisions that the methodological literature treats as routine are, in practice, sites where disciplinary evaluative norms leave their imprint.

Two findings complicate a simple domestication narrative. First, Psychology, not Political Science, occupies the most measurement-oriented position in the field. This challenges the widespread assumption that Political Science is the primary positivist adopter of computational text methods, and suggests that disciplinary positioning in the epistemic space reflects training traditions and evidentiary demands rather than stylized disciplinary reputations. Second, the within-discipline spreads reveal substantial internal heterogeneity, particularly in Psychology and Sociology. Domestication is not monolithic; it coexists with considerable within-field variation driven by subdisciplinary traditions, corpus characteristics, and the degree of co-authorship with computational specialists.

The negative interaction between algorithm complexity and year is the paper's most consequential finding. It indicates that the relationship between tool sophistication and documented calibration depth is not stable: as complex architectures diffuse to a broader user base, the calibration practices that characterized early adopters are not diffusing with them. This pattern is consistent with what we term epistemic drift, architectural convergence decoupled from substantive methodological engagement.

Several mechanisms may underlie this pattern, and our data cannot fully adjudicate among them. The drift may reflect a composition effect: as topic modeling diffuses beyond computationally specialized early adopters, the average

depth of engagement with the method's mathematical foundations may decline. It may reflect the scripting of practice by software packages (Akrich, 1992): bundled workflows like the `stm` R package allow researchers to produce reviewer-legible outputs by following default pipelines, reducing the need for, and perhaps the visibility of, deliberate calibration. It may also reflect the normative pressure of mimetic isomorphism (P. J. DiMaggio & Powell, 1983): under conditions of methodological uncertainty, adopting the most technically current architecture may be read by reviewers as a signal of competence, regardless of whether the architecture is calibrated to the specific analytical task.

What our data do establish is the structural pattern. The sophistication advantage of complex models is shrinking at a rate of approximately 0.095 index points per year (on a 0–5 scale), and this decoupling intensifies after controlling for discipline. The implication is that the field's increasing technical ambition is not being matched by a corresponding increase in the documented justificatory work that makes computational methods scientifically productive. Porter (1995) argued that standardized quantification gains authority as a "technology of distance", a mechanism that enables evaluation when reviewers lack localized trust. The decoupling we document suggests that this technology of distance may be operating asymmetrically: the adoption of technically prestigious architectures travels efficiently across disciplinary boundaries, but the discretionary calibration that would make those architectures locally meaningful does not.

The third finding, that between-discipline distances in the epistemic space are shrinking, gives the drift documented in RQ2 a field-level consequence. Disciplines are not only adopting the same tools; they are increasingly deploying those tools in similar ways. The 64% decline in centroid dispersion indicates that the epistemic diversity documented in the cross-sectional analysis is itself a historically specific configuration, not a stable feature of the field.

The convergence is not uniform, and the asymmetry is sociologically revealing. Psychology, Communication, and Political Science exhibit narrowing within-discipline spreads, consistent with consolidation around shared practices. Sociology and Interdisciplinary Social Sciences show widening spreads, their centroids move toward the field center, but individual papers become more dispersed. This pattern suggests that convergence operates through different pathways: in paradigmatically coherent disciplines, the entire field migrates; in fields with weaker internal consensus, the centroid shifts because some practitioners adopt measurement-oriented practices while others maintain interpretive positions, producing fragmentation rather than consolidation.

This asymmetry has implications for how we understand the relationship between disciplinary autonomy and methodological standardization. Sociology's internal fragmentation suggests that the field is not passively absorbing a foreign evaluative standard but is actively contested from within, with some practitioners embracing the measurement logic embedded in software defaults and others resisting it. The convergence we observe may therefore represent neither healthy maturation (in which case all disciplines would converge on shared calibration standards) nor simple colonization (in which case one discipline's practices would dominate). It may instead reflect the erosion of what we have termed

productive plasticity: the capacity of the boundary object to serve diverse epistemic communities precisely because each can adapt it to local evaluative needs (Star & Griesemer, 1989). As the center of the epistemic space comes to be defined by the default configurations of a small number of software packages rather than by any deliberate intellectual consensus, the diversity that enabled the method's cross-disciplinary diffusion is compressed.

5.1 Limitations

Several limitations qualify these findings. First, our data capture the documentary record of published practice, not practice itself; the gap between what researchers report and what they do may vary systematically across disciplines and journals. Second, the LLM-based extraction pipeline, while validated at $\kappa > 0.85$, may introduce measurement error that is correlated with article length, writing style, or disciplinary reporting conventions. Third, the early-period sample sizes for some disciplines are small (Sociology $n = 20$ in 2009–2019), and the yearly variance test ($n = 7$ year-points) is necessarily suggestive rather than conclusive. Fourth, our design cannot fully separate disciplinary effects from organizational confounds, corpus type, journal norms, and co-authorship patterns, that may covary with discipline. Finally, we operationalize convergence through centroid distance in a two-dimensional MCA space, and the results may be sensitive to the number of dimensions retained or to alternative geometric specifications. These limitations do not undermine the core patterns, but they constrain the strength of the causal inferences that can be drawn from them.

5.2 Implications

These findings speak to a question that extends well beyond topic modeling: how do the social sciences maintain evaluative autonomy when they absorb methods from technically more powerful but epistemologically distinct fields? The pattern we document, architectural convergence without calibration, accompanied by shrinking epistemic diversity, suggests that the threat to disciplinary autonomy comes not from the methods themselves but from the evaluative infrastructure that accompanies them: the default settings, reviewer expectations, and prestige hierarchies that make certain configurations of practice legible as rigorous while rendering others invisible.

The practical implication is not that disciplines should resist computational methods. It is that the social sciences need standards for what we might call interpretive transparency, reporting conventions that render the human–algorithm negotiation formally visible alongside algorithmic execution. Current transparency mandates overwhelmingly privilege codifiable dimensions of rigor: replication code, named metrics, and explicit preprocessing documentation. These are valuable, but they are not sufficient. A topic model whose K was selected by coherence score and whose labels were assigned by inspecting top words has satisfied the codifiable requirements; a model whose K was selected through iterative expert reading and whose labels were triangulated with ethnographic

knowledge of the empirical domain has often done more interpretive work, but that work is invisible under prevailing reporting formats. Rendering such work visible, through structured reporting of labeling rationale, validation procedures, and the reasoning behind K, would allow disciplines to demonstrate calibration without flattening it into a single engineering-derived metric. Espeland and Stevens (1998) remind us that commensuration always involves choices about which dimensions of a phenomenon to make visible; the design of reporting standards determines what forms of rigor the field can recognize.

As the social sciences confront the rapid diffusion of large language models and generative AI, tools whose internal operations are substantially more opaque than any topic model, the dynamics documented here offer an empirical foundation for anticipating how the next generation of computational tools may reshape the epistemic landscape. The question is not whether disciplines will adopt these tools; it is whether they will retain the evaluative autonomy to adapt them to the distinct standards of evidence that make disciplinary knowledge-making productive.

6 Conclusion

This study set out to examine how the social sciences domesticate a shared computational method, and what happens to disciplinary distinctiveness when software ecosystems exert standardizing pressure on that process. The evidence supports three interconnected conclusions.

First, topic modeling functions as a boundary object whose operational meaning is constituted through local translation. The same probabilistic architecture is adapted to serve fundamentally different epistemic purposes across disciplines, measurement in psychology, inferential scaffolding in political science, interpretive filtering in sociology, and descriptive mapping in communication. This adaptation is not limited to what researchers do with the model's output; it extends to the upstream decisions, K-selection, labeling, and validation, through which the model is rendered credible within each field. The implication is that discretionary methodological choices are not technical residuals to be standardized away; they are the observable traces of disciplinary warrant structures at work.

Second, the field is experiencing what we have termed epistemic drift: a widening gap between the complexity of the tools adopted and the depth of the documented calibration that accompanies them. Early adopters of complex architectures calibrated those tools more thoroughly than later adopters do. This pattern is consistent with a process in which the adoption of technically prestigious packages outpaces the transmission of the substantive expertise required to make those packages analytically productive. The drift is not a failure of any individual researcher; it is a structural feature of how software ecosystems interact with expanding user communities under conditions of evaluative uncertainty.

Third, the epistemic diversity documented in the cross-sectional analysis is not a permanent feature of the field. Between-discipline distances are shrinking,

and the convergence is driven not by the colonization of one field by another but by the retreat of formerly extreme positions toward a shared methodological center. This center is defined less by intellectual consensus than by the default configurations embedded in a small number of widely adopted software packages. The erosion of disciplinary distinctiveness, what we have called the loss of productive plasticity, matters because boundary objects derive their value precisely from the capacity to serve different communities differently. When that capacity is compressed, the method retains its mathematical identity but loses the evaluative diversity that enabled its diffusion across epistemic cultures.

These findings carry implications that extend beyond topic modeling. As the social sciences adopt large language models, generative AI, and other tools whose internal operations are substantially more opaque than any topic model, the dynamics documented here, architectural adoption outpacing calibration, software defaults overriding local evaluative norms, and disciplinary distinctiveness eroding under standardization pressure, are likely to intensify. The challenge shifts from whether to adopt computational methods to how to design reporting standards, training curricula, and review conventions that preserve the evaluative autonomy on which productive disciplinary knowledge-making depends.

Data and Code Availability Statement

The analytic corpus was constructed from bibliographic records indexed in the Web of Science Core Collection and from full-text articles acquired for research use. Because the underlying publications remain subject to publisher licensing and access restrictions, we do not redistribute the full-text corpus through the manuscript archive. The extraction schemas, prompt specifications, derived paper-level variables, and analysis code used to generate the reported findings and appendix tables will be made available in a public replication repository upon publication. Summary data necessary to reproduce the reported tables and figures will be included in that archive, together with documentation describing variable construction, recoding decisions, and the workflow linking extraction outputs to the final analyses.

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A Supplementary Statistical Tables

Table 2: Disciplinary association across implementation dimensions: bootstrapped, cluster-robust chi-square tests

Test	χ^2	df	p	Cramér's V	Bootstrap 95% CI
Epistemic Function (dominant)	55.1	20	$< .001$	0.150	[37.1, 112.1]
TM Link Type (dominant)	24.4	8	.002	0.141	[14.9, 53.7]
Analytical Family (dominant)	59.7	16	$< .001$	0.156	[42.5, 114.1]
Method Integration (dominant)	23.3	12	.026	0.113	[14.8, 55.7]
Algorithm	53.5	12	$< .001$	0.152	[39.3, 96.2]
K-Selection Strategy	38.9	16	.001	0.113	[31.9, 80.1]
Labeling Basis	36.0	16	.003	0.108	[26.7, 84.5]
Data Source	266.4	32	$< .001$	0.294	[232.4, 358.4]

Table 3: Distribution of active modalities included in the MCA ($N = 598$)

Variable	Modality	n	%
Epistemic Function	Descriptive Mapping	376	62.9
Epistemic Function	Data Reduction	131	21.9
Epistemic Function	Measurement Tool	91	15.2
TM Link Type	Structure	319	53.3
TM Link Type	Variables	158	26.4
TM Link Type	Filter	121	20.2
Analytical Family	Descriptive Presentation	366	61.2
Analytical Family	Computational Analysis	106	17.7
Analytical Family	Statistical Modeling	96	16.1
Analytical Family	Qualitative Engagement	30	5.0

B Methodological Appendix: Extraction Schema and Prompt Design

B.1 Structured Extraction Schema

The extraction pipeline was organized in two phases. Phase 1 targeted upstream technical decisions using the *Technique Bundle* (Methods + Results). Phase 2 targeted downstream analytical deployment using the *Downstream Task Bundle* (Results + Discussion + Conclusion). The raw Pydantic schemas used in the pipeline were converted into the publishable specification reported below.

Table 4: MCA modality coordinates on Dimensions 1 and 2

Variable	Modality	Dim 1	Dim 2
Epistemic Function	Data Reduction	-0.913	1.452
Epistemic Function	Descriptive Mapping	-0.138	-0.648
Epistemic Function	Measurement Tool	1.886	0.589
TM Link Type	Filter	-0.956	1.507
TM Link Type	Structure	-0.349	-0.744
TM Link Type	Variables	1.437	0.348
Analytical Family	Computational Analysis	-0.269	0.474
Analytical Family	Descriptive Presentation	-0.283	-0.455
Analytical Family	Qualitative Engagement	-1.276	2.061
Analytical Family	Statistical Modeling	1.774	0.567

Table 5: Supplementary discipline centroids and within-discipline spread in MCA space

Discipline	Dim 1	Dim 2	Trace
Sociology	-0.0914	0.1446	1.529
Political Science	0.0325	-0.0438	1.392
Communication	-0.1397	-0.1476	1.168
Psychology	0.2762	0.1124	1.670
Interdisciplinary Social Sciences	-0.1299	0.0176	1.036

Table 6: Logistic trend models for algorithm adoption and documented calibration components

Outcome	OR	95% CI	p
LDA use	0.904	[0.846, 0.967]	.0032
STM use	1.083	[1.002, 1.169]	.0433
BERTopic use	2.578	[1.560, 4.261]	.0002
Structural covariates	1.057	[0.976, 1.144]	.1733
Hyperparameters reported	1.006	[0.934, 1.083]	.8751
Named-metric K-selection	1.200	[1.119, 1.287]	$< .001$
Label validation	1.004	[0.912, 1.105]	.9381
Preprocessing reported	1.034	[0.955, 1.119]	.4150

Table 7: Distribution and component rates of the Methodological Sophistication Index

Component / Score	Rate / n	Value
Hyperparameters reported	27.7%	
Structural covariates	24.3%	
Named-metric K-selection	50.1%	
Preprocessing reported	79.6%	
Label validation	13.5%	
Score = 0	93	
Score = 1	164	
Score = 2	269	
Score = 3	177	
Score = 4	62	
Score = 5	4	
Mean	1.95	
SD	1.14	
Median	2	

Table 8: OLS models of methodological sophistication and the complexity × year decoupling effect

Variable	b	SE	p	Model
Intercept	1.6709	0.0749	$< .001$	Base
Algorithm complexity	0.6181	0.1407	$< .001$	Base
Centered year	0.0793	0.0212	.0002	Base
Complexity × centered year	-0.0899	0.0365	.0138	Base
Intercept	1.8101	0.0998	$< .001$	Full
Interdisciplinary Social Sciences	-0.2269	0.1206	.0599	Full
Political Science	-0.2080	0.1228	.0902	Full
Psychology	-0.0204	0.1121	.8560	Full
Sociology	-0.3945	0.1233	.0014	Full
Algorithm complexity	0.6639	0.1439	$< .001$	Full
Centered year	0.0750	0.0209	.0003	Full
Complexity × centered year	-0.0952	0.0366	.0093	Full

Note. Base model: $R^2 = 0.0514$, $N = 707$. Full model: $R^2 = 0.0683$, adjusted $R^2 = 0.0590$, $N = 707$.

Table 9: Discipline centroids by period and summary indicators of convergence in MCA space

Discipline	Period	n	Dim 1	Dim 2
Sociology	2009–2019	20	-0.210	0.009
Sociology	2020–2024	69	-0.057	0.184
Political Science	2009–2019	29	-0.003	0.078
Political Science	2020–2024	98	0.043	-0.080
Communication	2009–2019	28	-0.103	-0.004
Communication	2020–2024	135	-0.147	-0.177
Psychology	2009–2019	23	0.593	0.275
Psychology	2020–2024	113	0.212	0.079
Interdisciplinary Social Sciences	2009–2019	11	0.149	-0.153
Interdisciplinary Social Sciences	2020–2024	72	-0.172	0.044

Summary indicators. Converging pairs: 7/10. Mean pairwise distance: $0.426 \rightarrow 0.287$ (-32.6%). Centroid dispersion: $0.1223 \rightarrow 0.0445$ (-64.0%).

Table 10: Between-discipline variance in MCA space by year and within-discipline spread by period

Year / Discipline	Variance / Early trace	n early	Late trace / variance	n late
2018	0.7371	20		
2019	0.3009	40		
2020	0.0936	44		
2021	0.1413	71		
2022	0.0986	96		
2023	0.1133	110		
2024	0.0790	166		
Sociology	1.441	20	1.563	69
Political Science	1.551	29	1.354	98
Communication	1.530	28	1.099	135
Psychology	1.922	23	1.604	113
Interdisciplinary Social Sciences	0.843	11	1.058	72

Note. Correlation between year and between-discipline variance: $r = -0.758$, $p = .0485$.

Table 11: OLS regression of distance from the field centroid on year, discipline, and their interaction

Variable	b	SE	p
Intercept	1.0950	0.0685	$< .001$
Interdisciplinary Social Sciences	-0.2268	0.0950	.0169
Political Science	0.0356	0.0885	.6879
Psychology	0.2045	0.1194	.0869
Sociology	-0.0210	0.1112	.8501
Centered year	-0.0364	0.0175	.0371
Year × Interdisciplinary Social Sciences	0.0561	0.0262	.0324
Year × Political Science	0.0125	0.0234	.5953
Year × Psychology	0.0067	0.0311	.8288
Year × Sociology	0.0633	0.0313	.0431

Note. $R^2 = 0.0512$, $N = 580$.

Table 12: Early-to-late change in measurement-oriented deployment by discipline

Discipline	Indicator	Early (%)	Late (%)	Change (pp)
Sociology	Measurement Tool	10.0	14.5	+4.5
Sociology	Variables (TM Link)	15.0	27.5	+12.5
Political Science	Measurement Tool	20.7	17.2	-3.5
Political Science	Variables (TM Link)	20.7	30.3	+9.6
Communication	Measurement Tool	10.7	8.8	-1.9
Communication	Variables (TM Link)	21.4	15.4	-6.0
Psychology	Measurement Tool	39.1	20.9	-18.2
Psychology	Variables (TM Link)	60.9	33.9	-27.0
Interdisciplinary Social Sciences	Measurement Tool	7.7	8.9	+1.2
Interdisciplinary Social Sciences	Variables (TM Link)	61.5	30.4	-31.1

Table 13: Phase 1 extraction schema: upstream technical variables

Schema block	Core fields	Analytic role
DataExtraction	Data source, corpus size, unit of analysis, preprocessing	Identifies the textual object modeled and the degree to which preprocessing is made visible.
ModelExtraction	Algorithm, software, final K, metadata in model, hyperparameter reporting	Captures the architecture selected and whether the model-fitting process is documented in sufficient detail to evaluate calibration.
KSelectionExtraction	Selection strategy, metrics used, human judgment, verbatim evidence	Operationalizes how authors justify the final number of topics and whether they rely on metrics, human interpretation, or both.
LabelingExtraction	Labeling basis, labeling process, label validation, verbatim evidence	Captures how statistical output is translated into substantive categories and whether that translation is formally validated.
ReproducibilityExtraction	Code and/or data sharing	Measures the visibility of replication materials in the published record.

Table 14: Phase 2 extraction schema: downstream analytical variables

Schema block	Core fields	Analytic role
DownstreamComponent	TM link type, epistemic function, topic-model position, topic-output form, analytical family, inferential claim, validation approach, methodological integration	Identifies each distinct way in which topic-model output is inserted into substantive analysis.
DownstreamExtractionLLM	Presence of downstream analysis, component list, global description	Determines whether topic-model output is interpreted, used as input, or merely reported without further analysis.
RescueKSelection / RescueLabeling	Supplemental extraction from Results/Discussion	Recovers K-selection and labeling evidence that appears outside formal methods sections.
DownstreamExtractionFull	Rollups of epistemic function, link type, output form, inferential strength, and integration	Aggregates multi-component downstream use to the paper level for cross-tabulation and geometric analysis.

B.2 Prompt Design in Publishable Form

The prompts were designed around five principles. First, extraction was restricted to the *current study* only; literature-review references and generic methodological exposition were excluded. Second, topic modeling was treated as executed only if the article contained explicit execution signals (e.g., "we fit"), concrete model outputs (e.g., topic-word lists, stated topic numbers), or unmistakable references to a finalized topic model. Third, evidence fields were required to contain verbatim quotations from the article. Fourth, missing-value mapping was conservative: fields defaulted to `NOT_REPORTED` unless explicit textual support existed. Fifth, evidence gating rules forced internal consistency. For example, a reported final K required a supporting quote; otherwise the field was reset to missing.

For downstream extraction, component creation was governed by a *topic-model link gate*. A downstream component was created only if the excerpt explicitly connected topic-model output (e.g., topics, topic proportions, assignments) to an analytic action (e.g., interpretation, comparison, regression, filtering). If a paper merely listed topics without interpretation or use, it was coded as having no downstream analysis. If the excerpt appeared truncated and lacked topic-output cues, it was coded as not reported.

B.3 Operational Rules

- **Own-study rule:** only actions clearly attributable to the focal article were coded.
- **Precision-first coding:** no practice was coded as present without explicit evidence.
- **Evidence quotations:** all evidence fields required verbatim quotations from the article.
- **Execution gate:** topic modeling was treated as executed only when explicit action, concrete outputs, or unambiguous topic-count language appeared.
- **Rescue logic:** K-selection and labeling evidence could be recovered from Results/Discussion when absent from formal methods sections.

B.4 Illustrative Field Mapping

To make the pipeline reproducible in prose form, Table 15 summarizes the principal coding translations from article text to analytic variable.

Table 15: Illustrative field-mapping rules for the extraction pipeline

Observed article statement	Coding consequence
"We fit a 30-topic STM with party as a prevalence covariate"	Algorithm = STM; $K = 30$; Metadata in model = Structural Covariates.
"We selected $K = 30$ based on perplexity and interpretability"	K-selection = Named Metric Plus Human; metric used = Perplexity.
"Two coders independently labeled topics; $\kappa = 0.82$"	Labeling process = Structured Multi-Coder; Label validated = Yes.
"Topic 3 captures health care reform" with no description of labeling procedure	Interpretation present, but labeling basis remains NOT_REPORTED.
`scikit-learn` mentioned only for preprocessing	Software for topic modeling remains NOT_REPORTED.
Topic list only, with no interpretation or analytic use	Downstream analysis = No.